

ES -116

Principles and Applications of Electrical Engineering

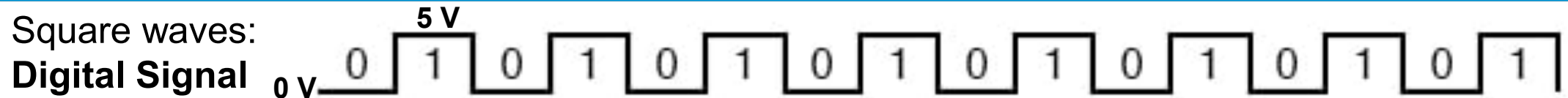
Semester II, 2025-26

Interfacing the Analog and Digital Worlds

Reference book: Chapter on “Interfacing with the Analog World” from Digital Systems Book by Tocci

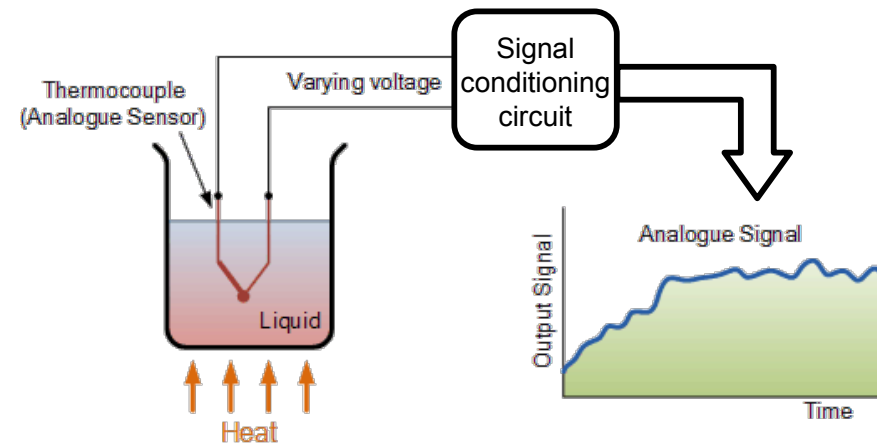
Recap

Square waves:

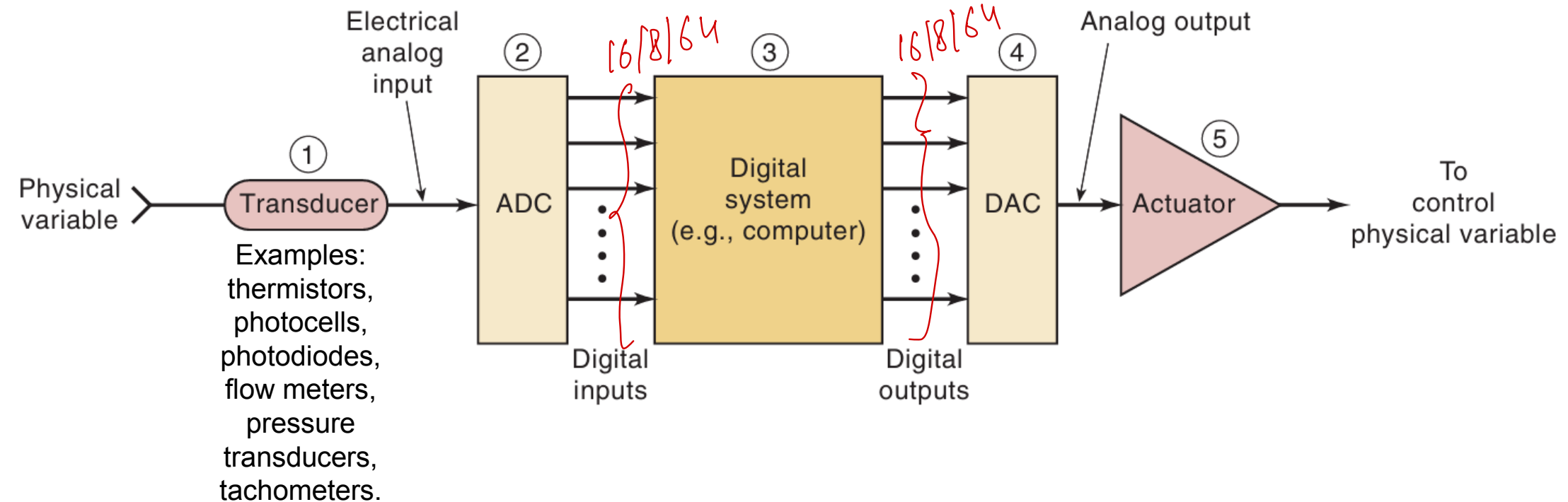


- A **digital** quantity has a value that is specified as one of two possibilities, such as 0 or 1, LOW or HIGH or true or false.
- In practice, a digital quantity such as a voltage may actually have a value that is anywhere within specified ranges.
- For example, 0 V to 0.8 V $\rightarrow$ logic 0 and 2 V to 5 V $\rightarrow$ logic 1
- It means that any voltage falling in the range from 0 to 0.8 V is given the digital value 0, and any voltage in the range 2 to 5 V is assigned the digital value 1.

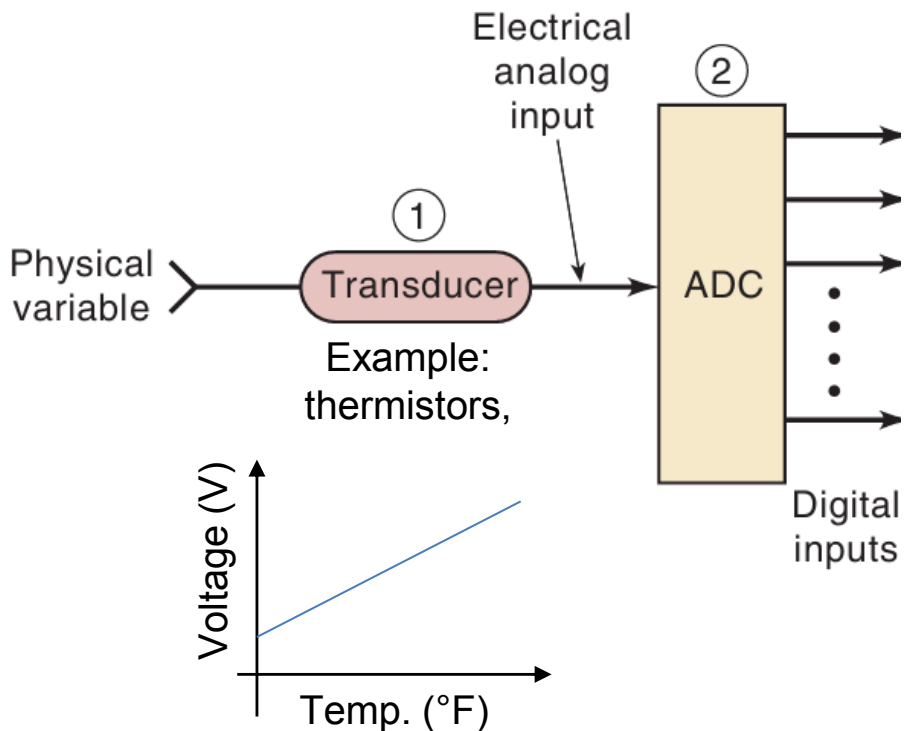
- By contrast, an **analog** quantity can take on any value over a continuous range of values and, most important, its exact value is significant.
- For example, the output of an analog temperature-to-voltage converter might be measured as 2.76 V, which may represent a specific temperature of 27.6°C.
- If the voltage were measured as something different, such as 2.34 V or 3.78 V, this would represent a completely different temperature.



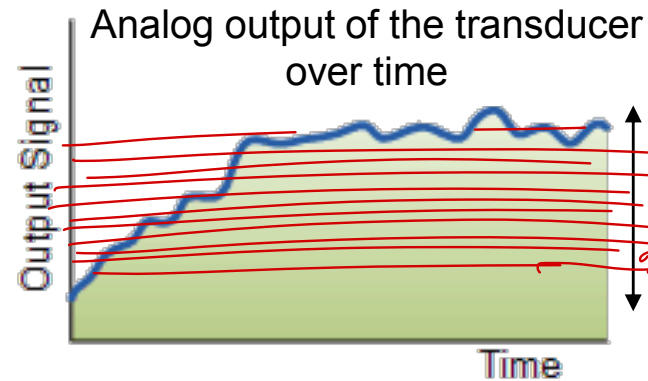
A Representative Block Diagram of a System



Role of ADC



Example: Voltage vs Temperature plot for a thermistor, 1°F produces a 10 mV output. We want to monitor temperatures between 80°F and 150°F. Voltage range = 700 mV.



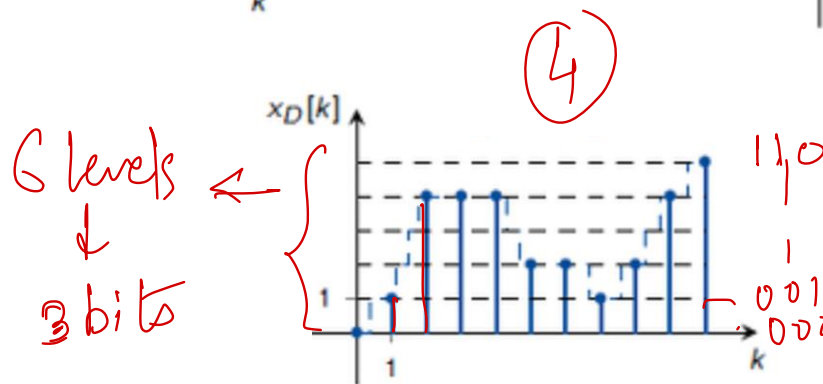
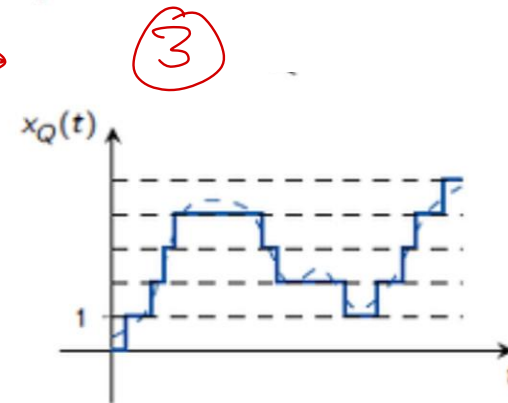
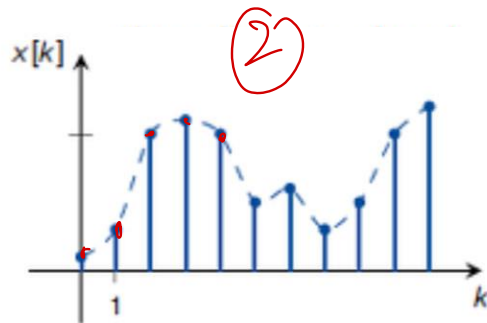
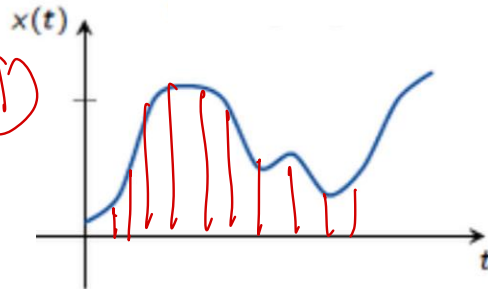
bit
 0
 1
 $\frac{FSR}{2^n} = LSB$

What do I need to decide before converting it into a digital output if I want a precision of 1°F?

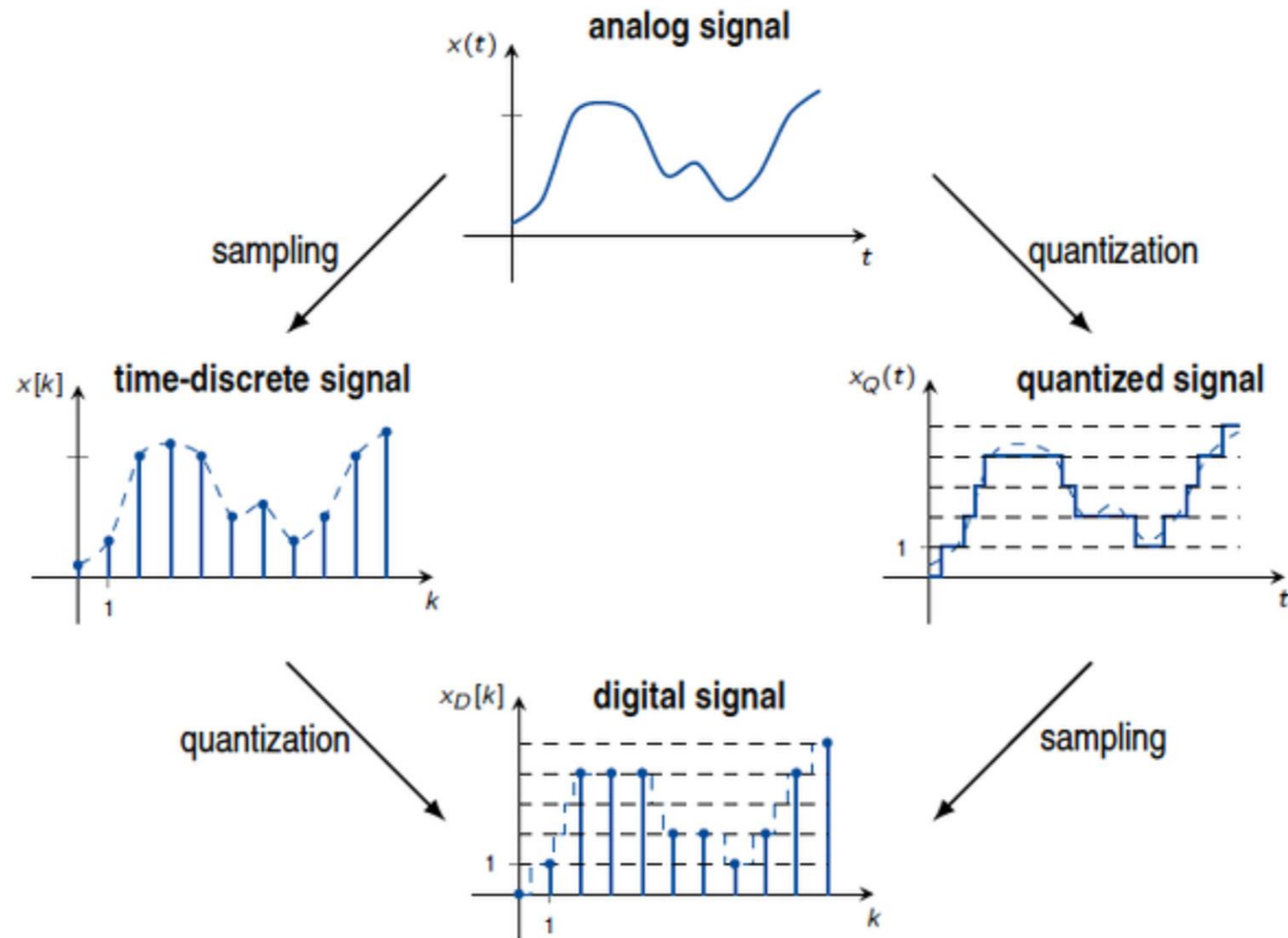
$FSR = 700mV$, Digital bit(n)
 Number of levels
 $= 2^n$

Match the four graphs to their correct terms

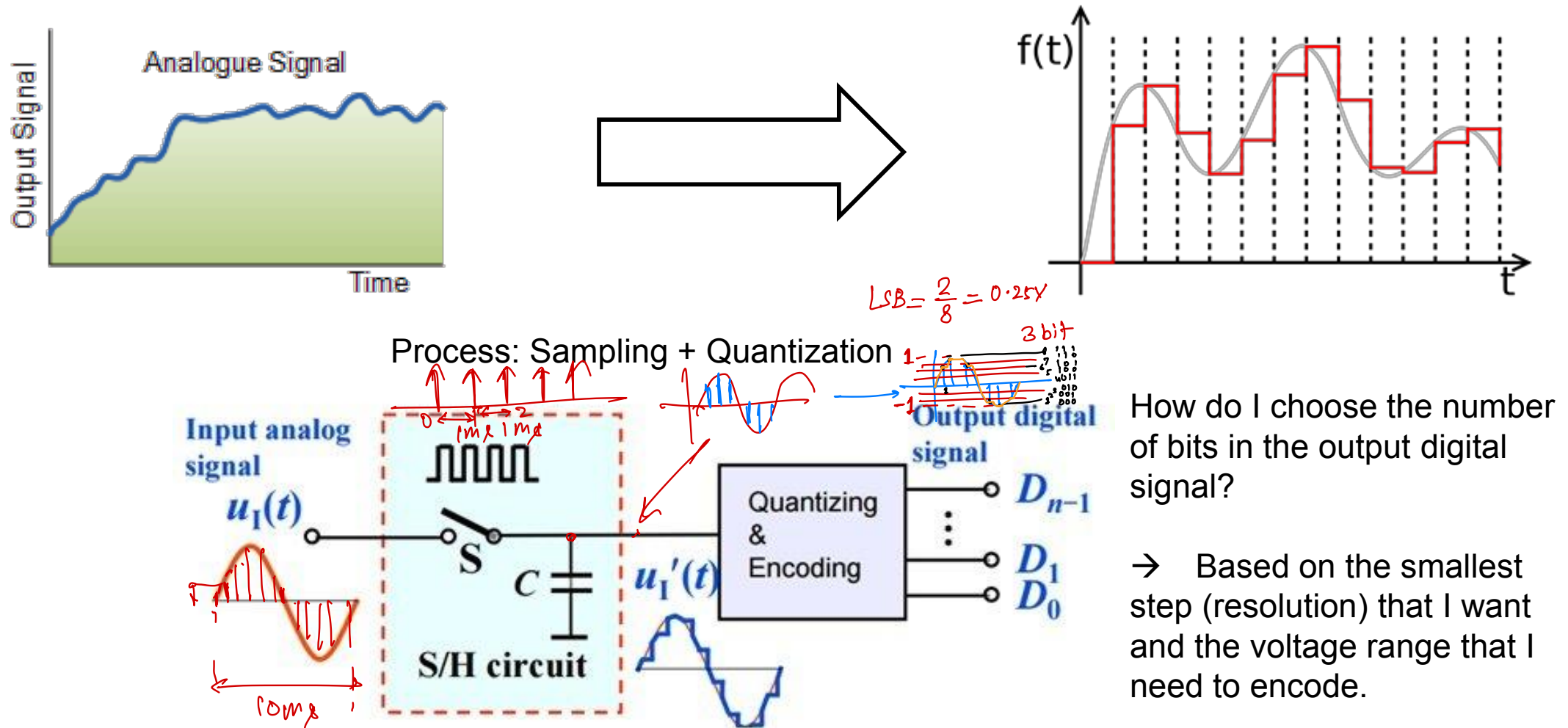
- Pure Analog (1)
- Sampled (Discrete-Time) (2)
- Quantized (Discrete-Amplitude) (3)
- True Digital (4)



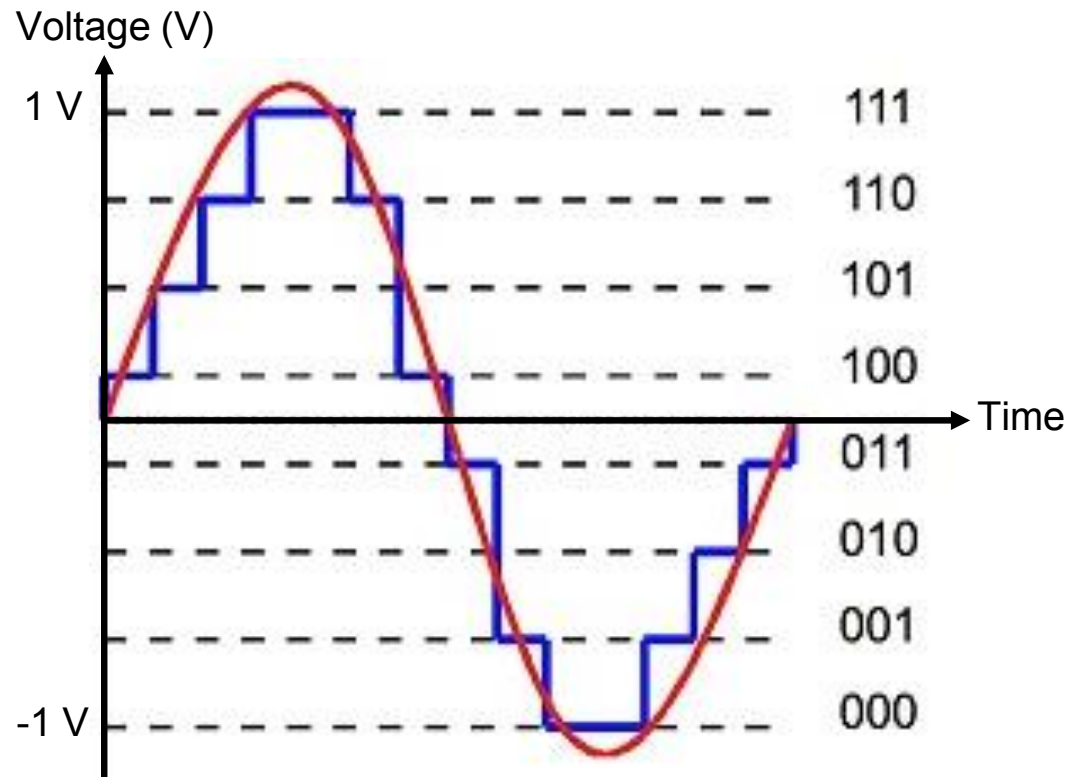
Classification of Signals in Time and Amplitude



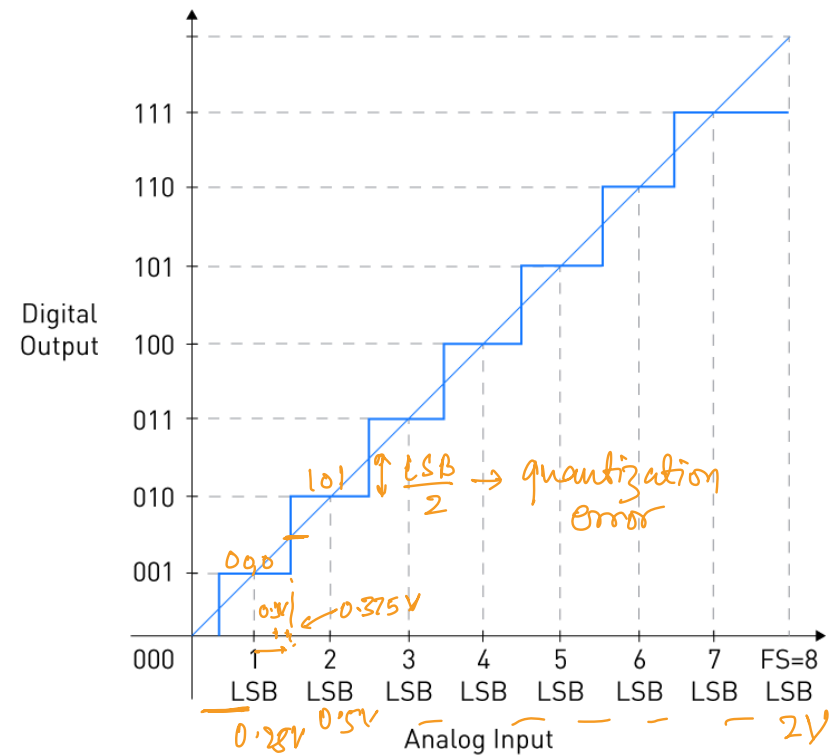
Analog to Digital Conversion



Analog to Digital Conversion



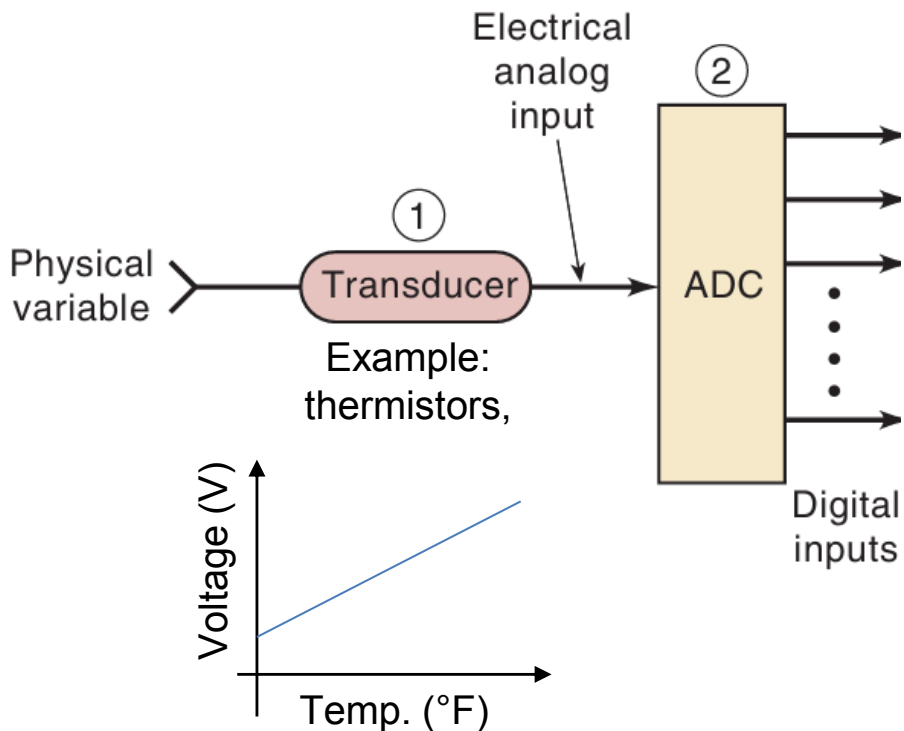
$$LSB = \frac{FSR}{2^n}$$



Q. For coding Full Scale voltage Range (FSR) of 2 V into 3 bit (n) Digital output, what will be the smallest step size?

$$\Delta = \frac{2V}{8} = 0.25V$$

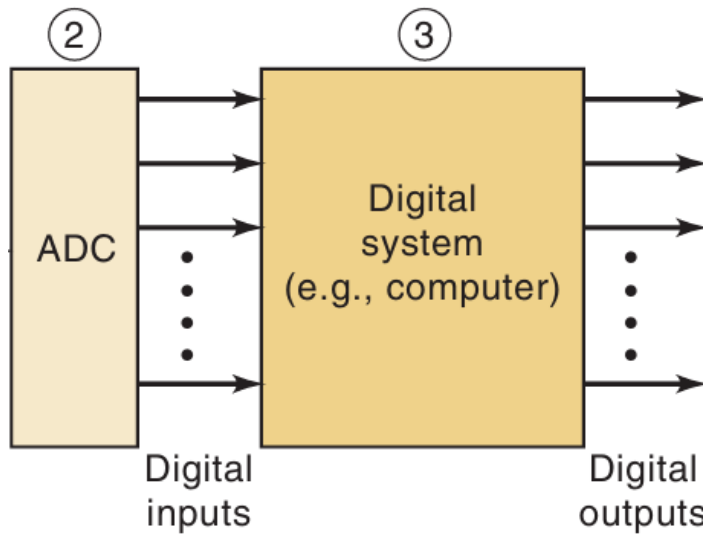
Expectations from ADC



Example: Voltage vs Temperature plot for a thermistor, 1°F produces a 10 mV output. We want to monitor temperatures between 80°F and 150°F. Voltage range = 700 mV.

- The transducer's electrical analog output serves as the analog input to the analog-to-digital converter (ADC).
- The ADC converts this analog input to a digital output.
- This digital output consists of a number of bits that represent the value of the analog input.
- For example, the ADC might convert the transducer's 800 to 1500 mV analog values to binary values ranging from 01010000 (80) to 10010110 (150).
- Note that the binary output from the ADC is proportional to the analog input voltage so that each unit of the digital output represents **10 mV**.

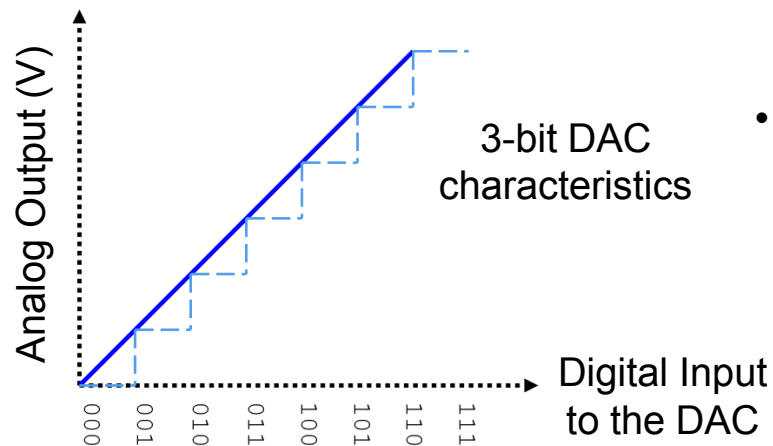
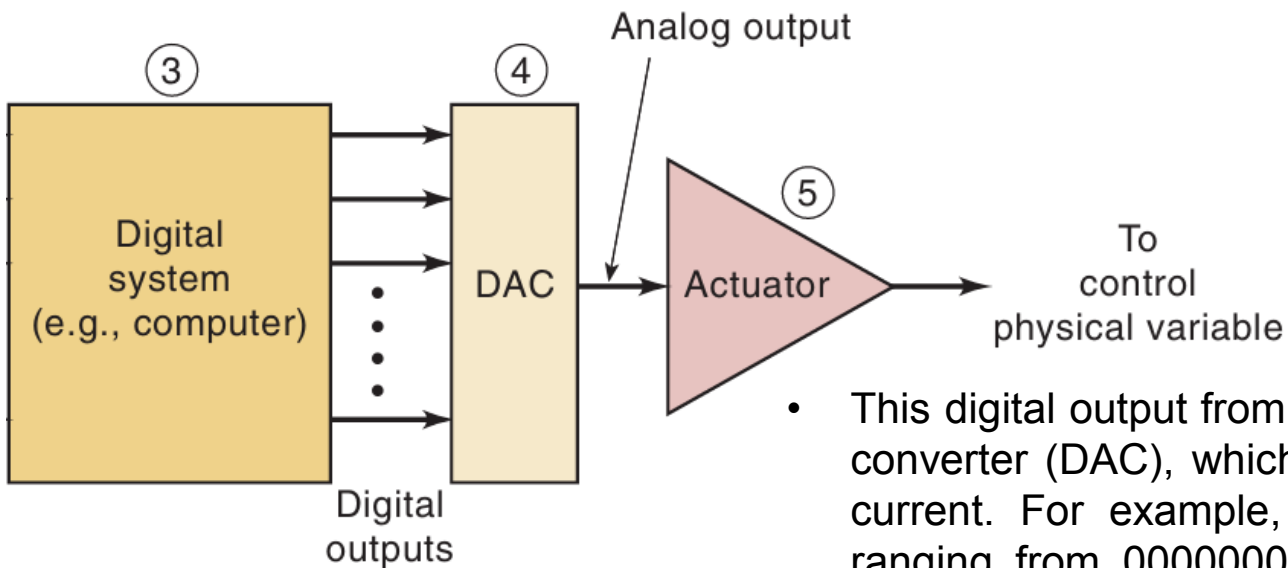
Computer processing in Digital Domain



- The digital representation of the process variable is transmitted from the ADC to the digital computer, which stores the digital value in its memory and processes it according to a program of instructions that it is executing.
- The program might perform calculations or other operations on this digital representation of temperature to come up with a digital output that will eventually be used to control the temperature.

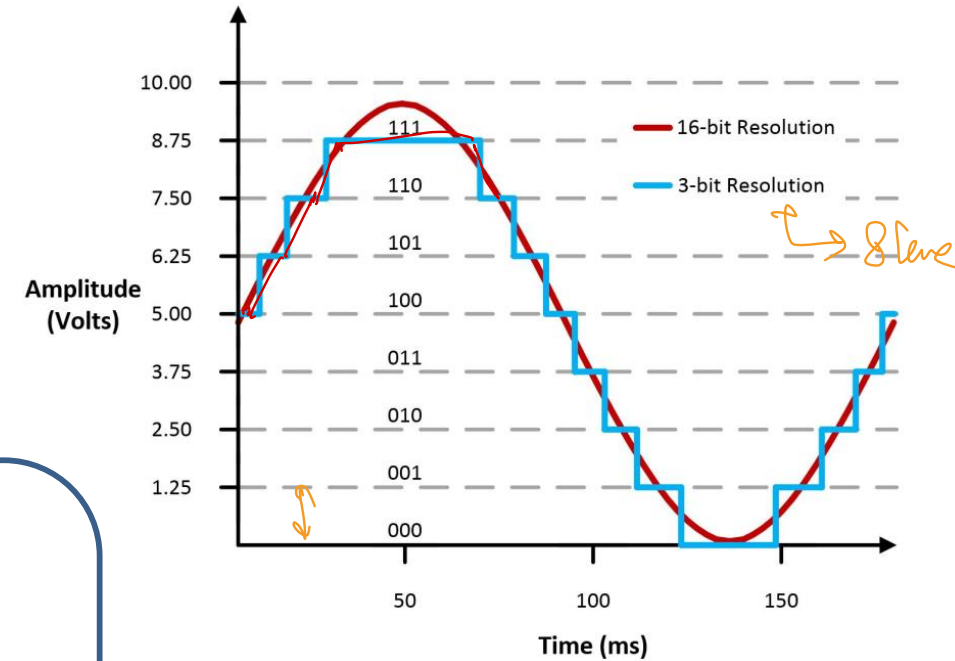
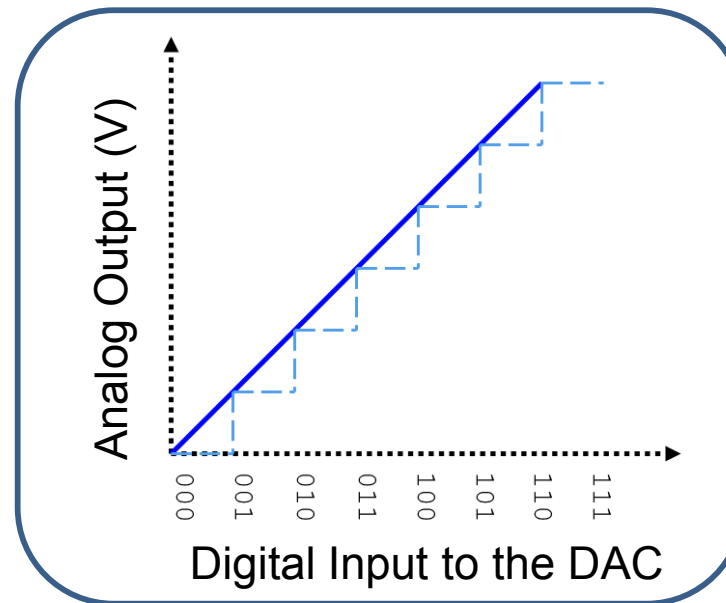
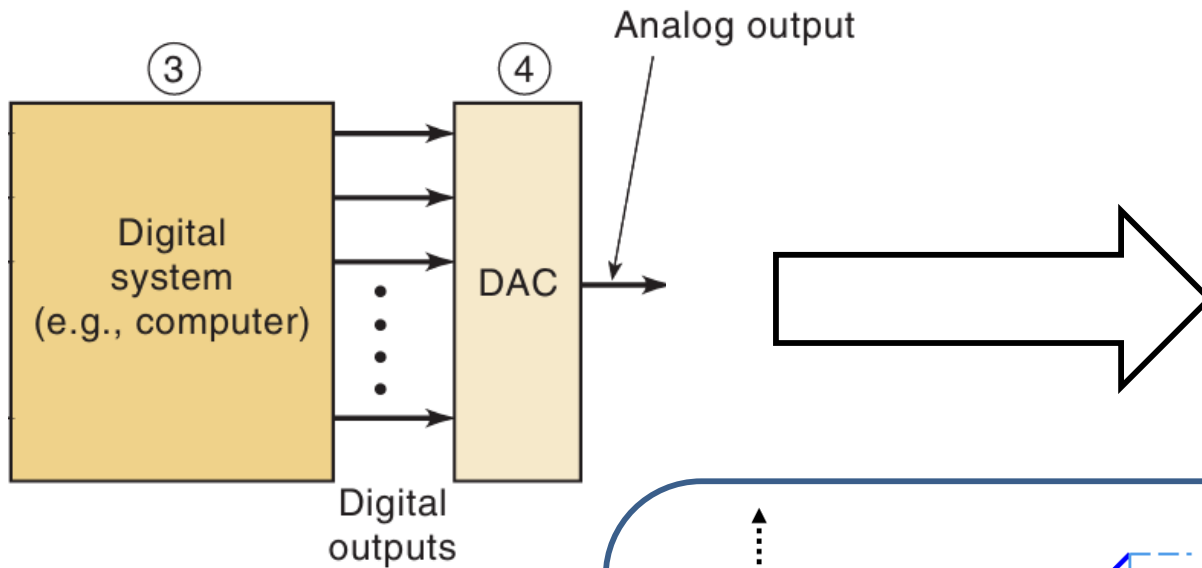
Digital values are also stored in terms of Voltages within the computer!

Expectations from DAC



- This digital output from the computer is connected to a digital-to-analog converter (DAC), which converts it to a proportional analog voltage or current. For example, the computer might produce a digital output ranging from 00000000 to 11111111, which the DAC converts to a voltage ranging from 0 to 10 V.
- The analog signal from the DAC is often connected to some device or circuit that serves as an actuator to control the physical variable. For the temperature example, the actuator might be an electrically controlled valve that regulates the flow of hot water into the tank in accordance with the analog voltage from the DAC. The flow rate would vary in proportion to this analog voltage, with 0 V producing no flow and 10 V producing the maximum flow.

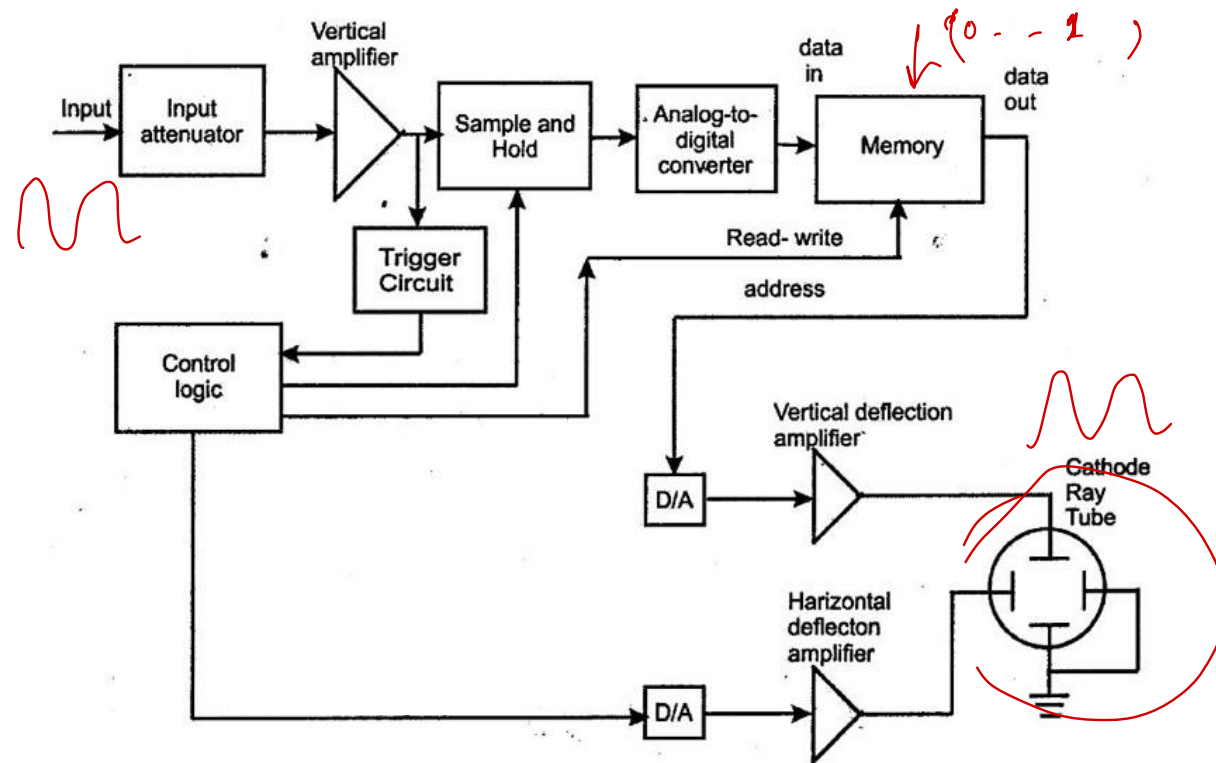
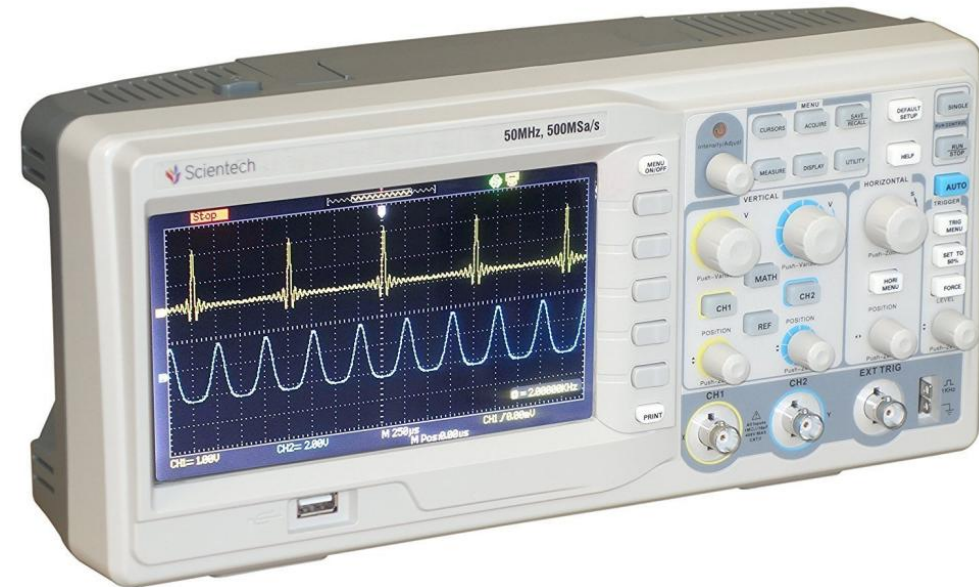
DAC output



Impact of resolution on the sinusoid output

$$\frac{10V}{8} = 1.25V$$

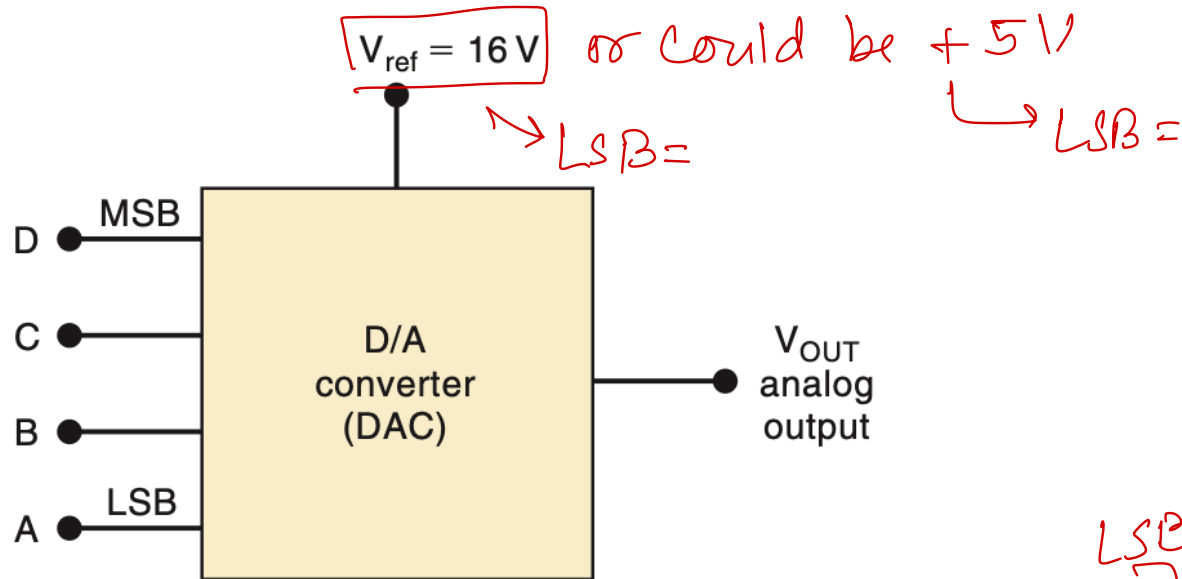
A fine example



Block diagram of a Digital Storage Oscilloscope (DSO)

What does the stop button do on your DSO?

Let's build a DAC circuit!



D	C	B	A	V_{OUT}	
0	0	0	0	0	Volts
0	0	0	1	1	
0	0	1	0	2	
0	0	1	1	3	
0	1	0	0	4	
0	1	0	1	5	
0	1	1	0	6	
0	1	1	1	7	
1	0	0	0	8	
1	0	0	1	9	
1	0	1	0	10	
1	0	1	1	11	
1	1	0	0	12	
1	1	0	1	13	
1	1	1	0	14	
1	1	1	1	15	Volts

2^3	2^2	2^1	2^0
1	$\frac{1}{2}$	$\frac{1}{4}$	$\frac{1}{8}$

Can you try to write the relation of V_{OUT} in terms D, C, B, A, and V_{ref} ?

$$\begin{aligned}
 V_{out} &= (2^3 \cdot D + 2^2 \cdot C + 2^1 \cdot B + 2^0 \cdot A) \times \frac{V_{ref}}{2^4} \\
 &= \frac{V_{ref}}{2} \left(D + \frac{C}{2} + \frac{B}{4} + \frac{A}{8} \right)
 \end{aligned}$$

or

How to build this functionality using OP-Amp?

$$V_{out} = \frac{V_{ref}}{2} * \left[D + \frac{C}{2} + \frac{B}{4} + \frac{A}{8} \right] \text{ or } \frac{V_{ref}}{16} * [8D + 4C + 2B + A]$$

another possibility

$V_D = V_{ref} \times D$

$\left\{ \begin{array}{l} '1' \rightarrow +V_{ref} \\ '0' \rightarrow 0 \end{array} \right\}$

$\left\{ \begin{array}{l} '1' \rightarrow -V_{ref} \\ '0' \rightarrow 0 \end{array} \right\}$

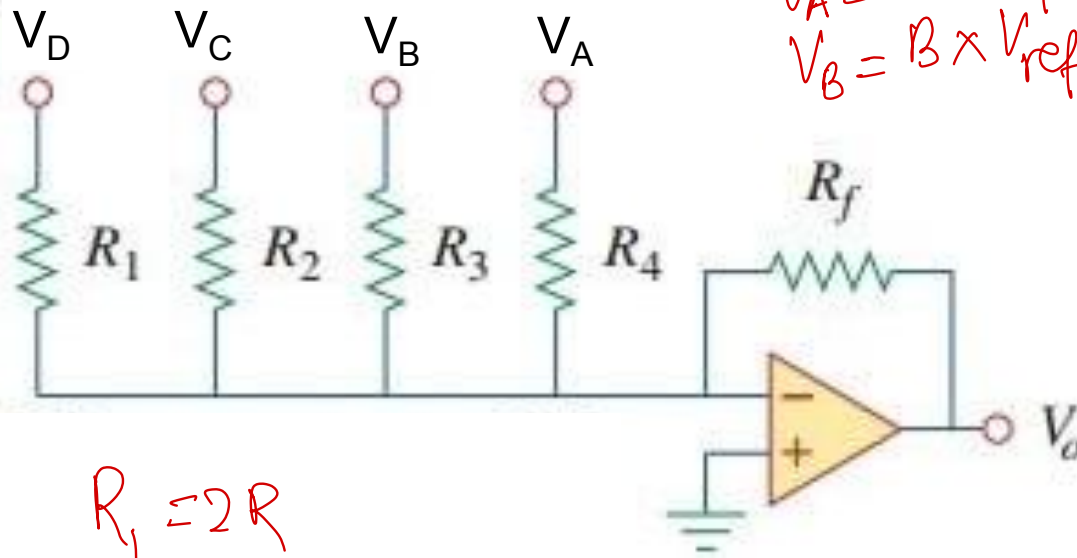
logical $\rightarrow$ physical

4-bit DAC circuit with Op-Amp

What should be the values of R_{1-4} if $R_f = R$ and Full-scale Output = -15 V

$$V_{out} = \frac{V_{ref}}{2} * \left[D + \frac{C}{2} + \frac{B}{4} + \frac{A}{8} \right]$$

$$\begin{aligned} V_A &= A \times V_{ref} & V_C &= C \times V_{ref} \\ V_B &= B \times V_{ref} & V_D &= D \times V_{ref} \end{aligned}$$

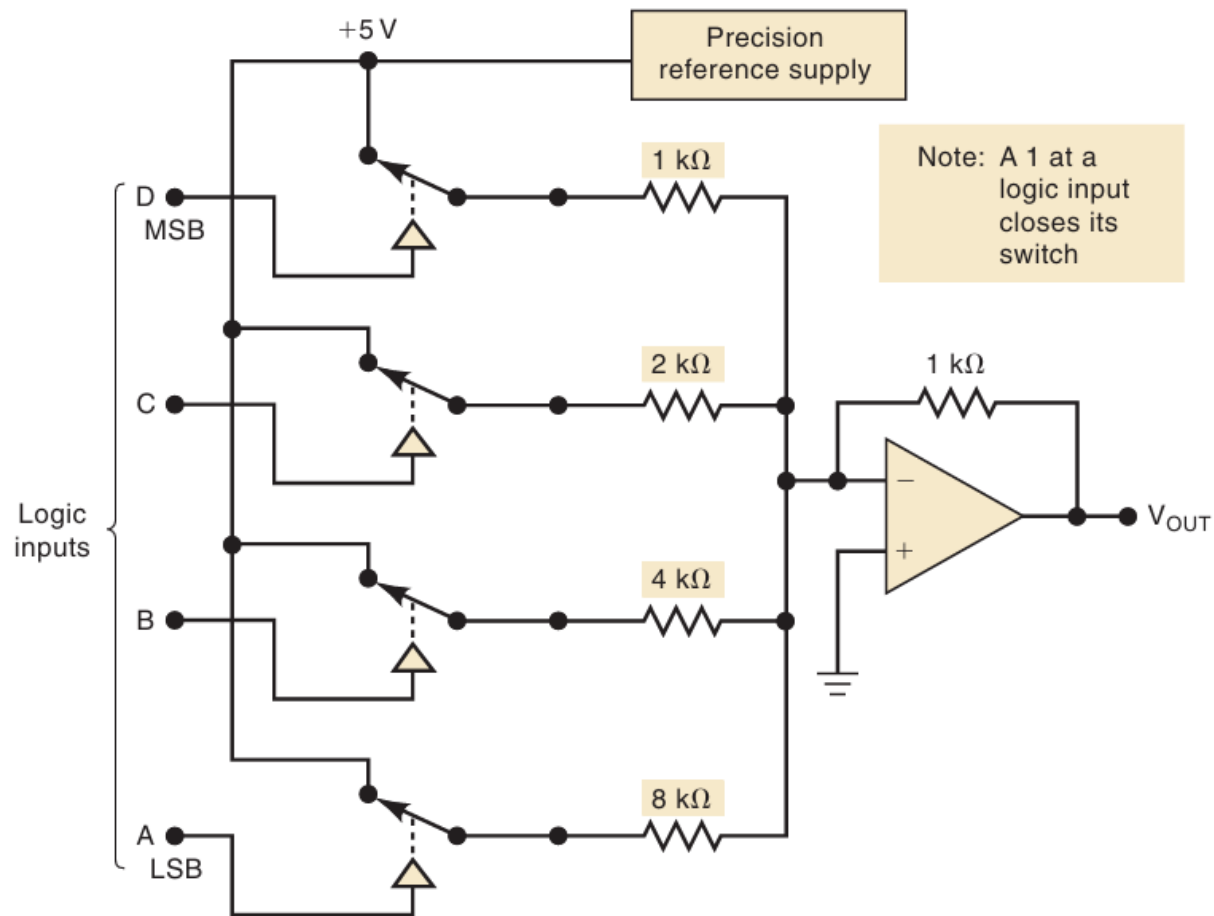


$$V_o = - \left(\frac{R_f V_D}{R_1} + \frac{R_f V_C}{R_2} + \frac{R_f V_B}{R_3} + \frac{R_f V_A}{R_4} \right)$$

$$= -V_{ref} \cdot R \left[\frac{D}{2R} + \frac{C}{4R} + \frac{B}{8R} + \frac{A}{16R} \right]$$

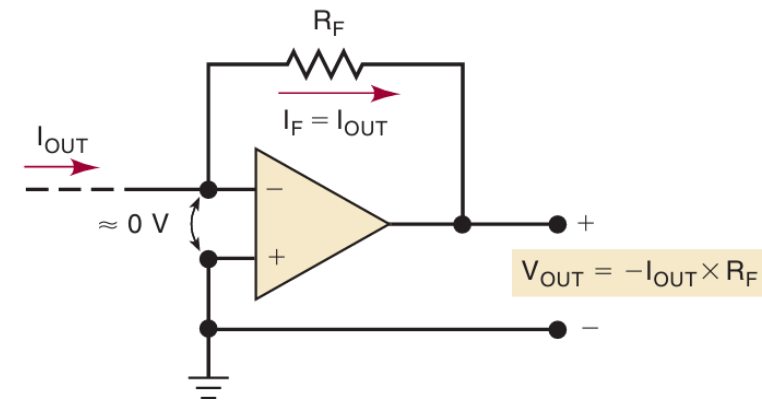
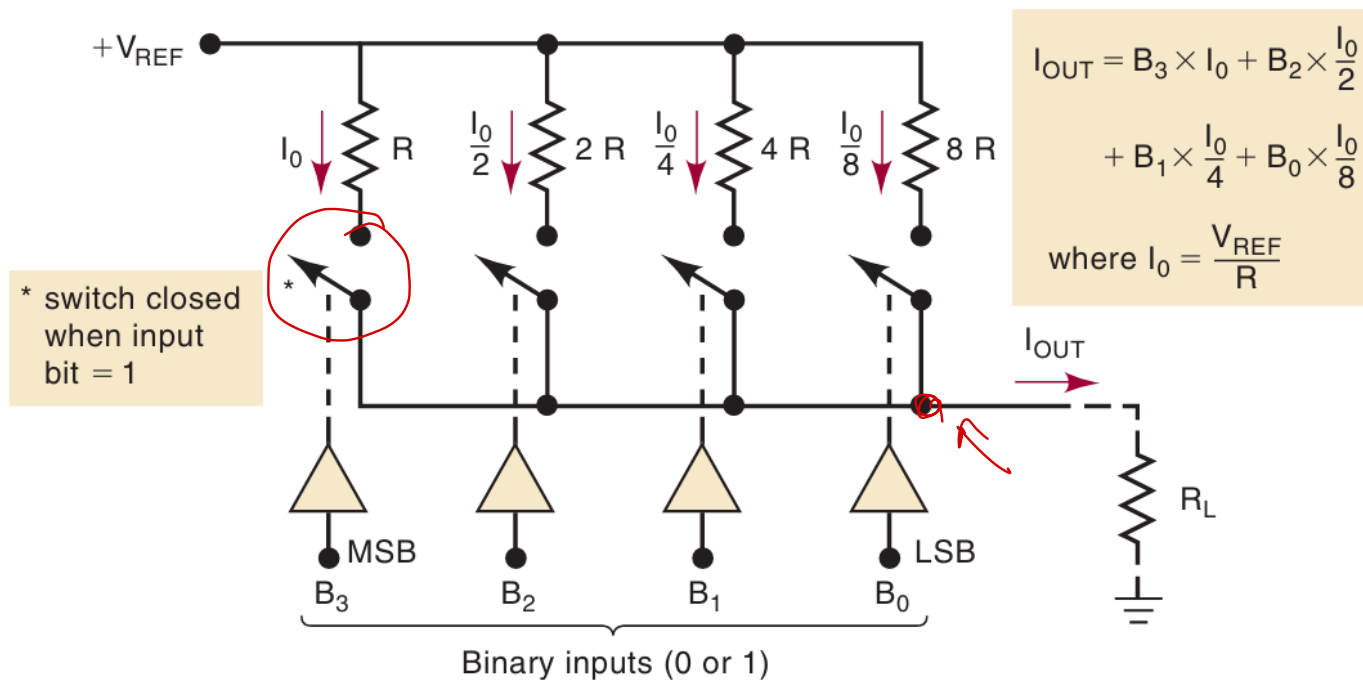
$$= - \frac{V_{ref}}{2} \left[D + \frac{C}{2} + \frac{B}{4} + \frac{A}{8} \right]$$

4-bit DAC circuit with Op-Amp

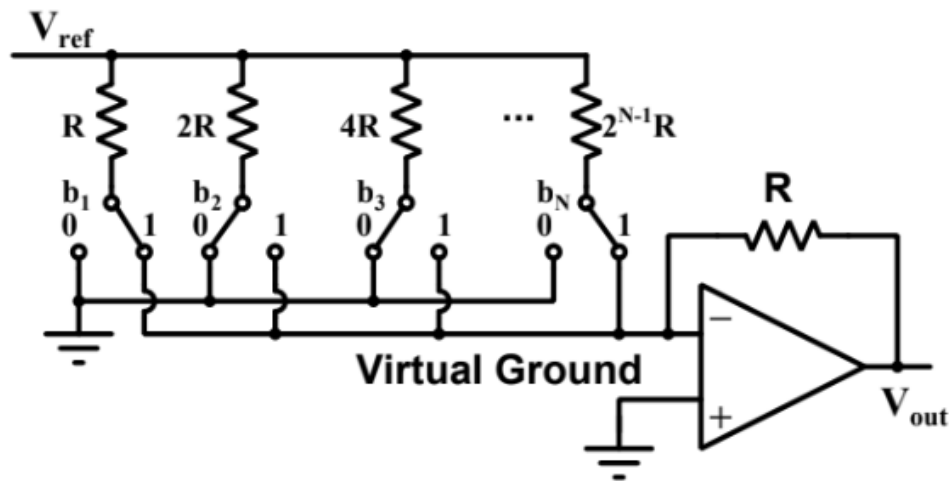


Input code				V _{OUT} (volts)
D	C	B	A	
0	0	0	0	0
0	0	0	1	-0.625 ← LSB
0	0	1	0	-1.250
0	0	1	1	-1.875
0	1	0	0	-2.500
0	1	0	1	-3.125
0	1	1	0	-3.750
0	1	1	1	-4.375
1	0	0	0	-5.000
1	0	0	1	-5.625
1	0	1	0	-6.250
1	0	1	1	-6.875
1	1	0	0	-7.500
1	1	0	1	-8.125
1	1	1	0	-8.750
1	1	1	1	-9.375 ← Full-scale

Using current to voltage conversion



Binary Weighted Resistors

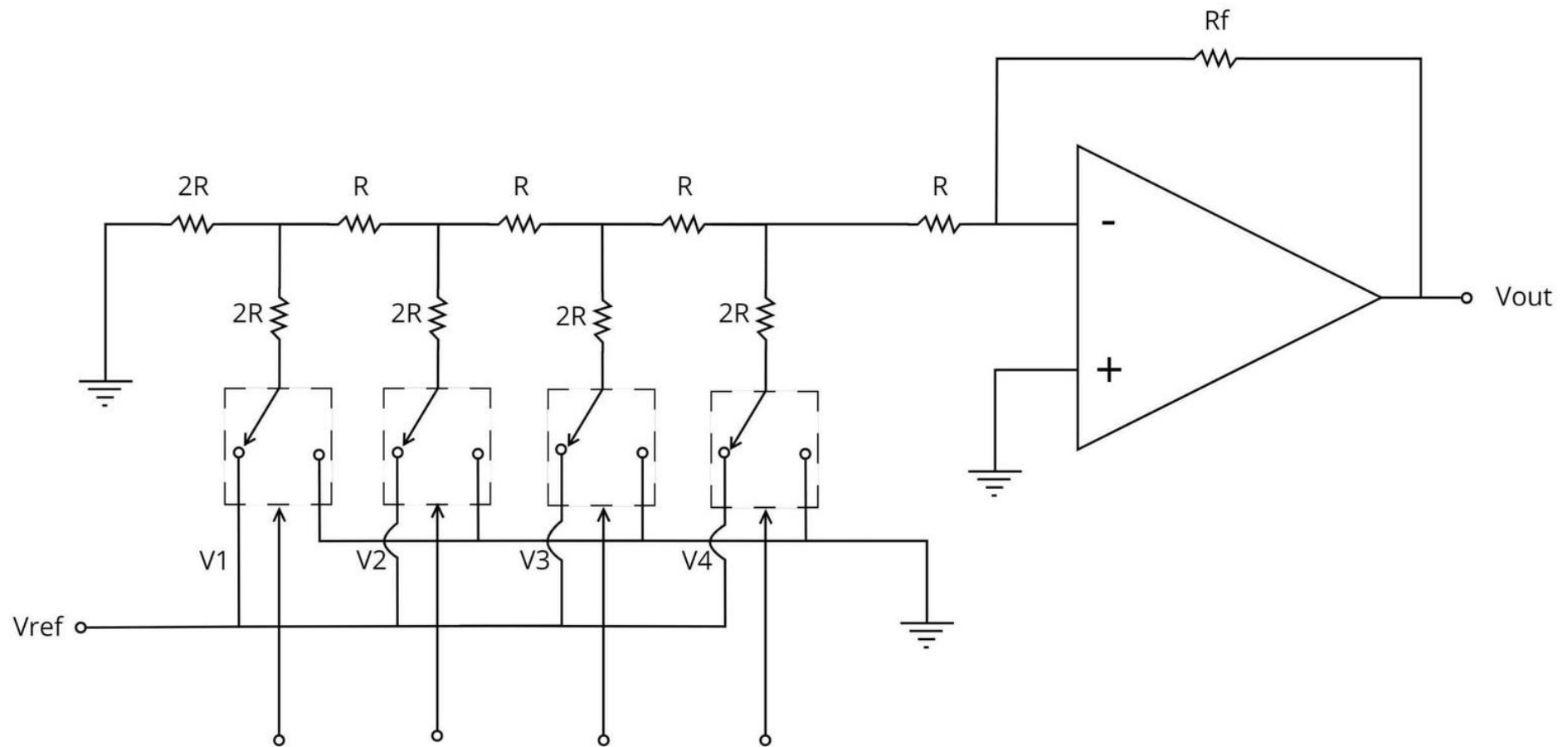


- Each bit corresponds to a switch.
- Current goes to ground if bit is 0.
- Currents are added into summing node if bits are 1.
- The opamp converts current back to voltage.

$$V_{out} = -R \sum_{i=1}^N \frac{V_{ref}}{2^i R} b_i = -\frac{V_{ref}}{2^N} \sum_{i=1}^N 2^{N-i} b_i$$

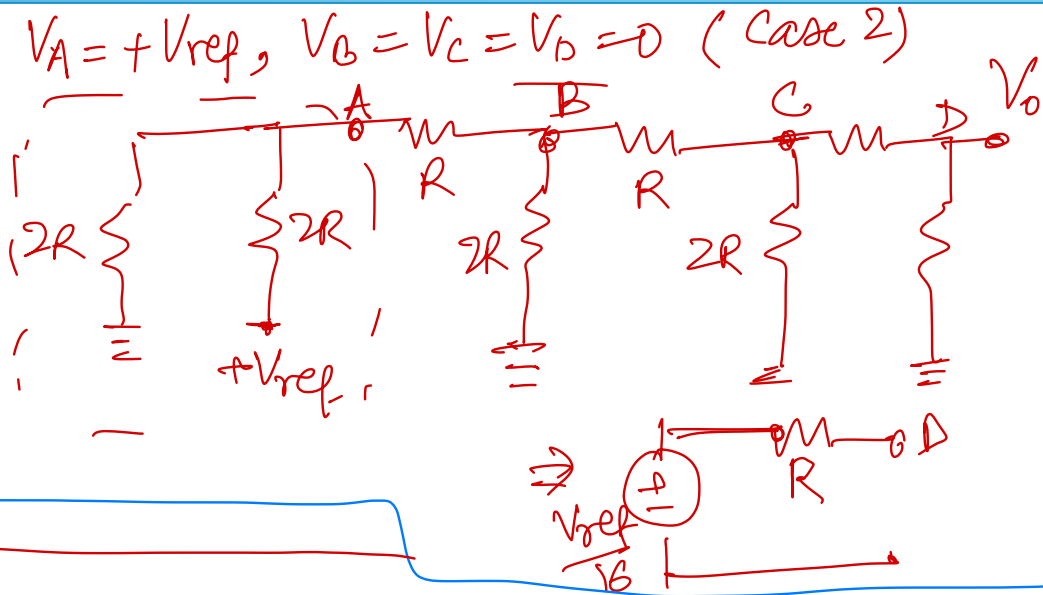
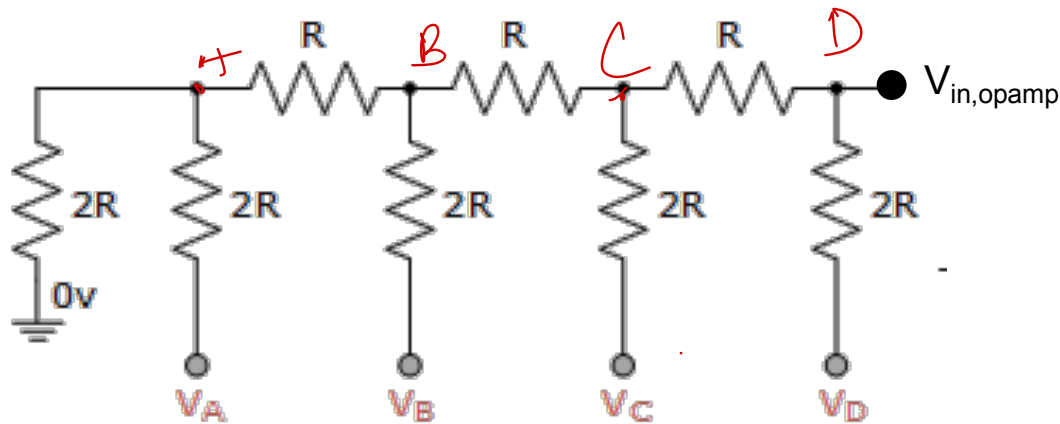
What is the downside of this circuit? [Hint: Does it scale well with number of bits]

Another implementation (Lab 10)



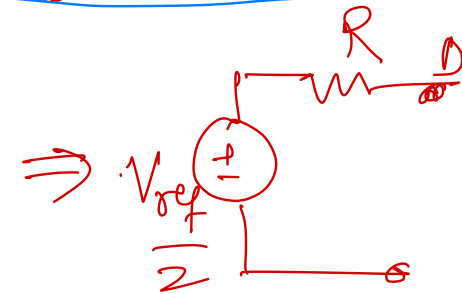
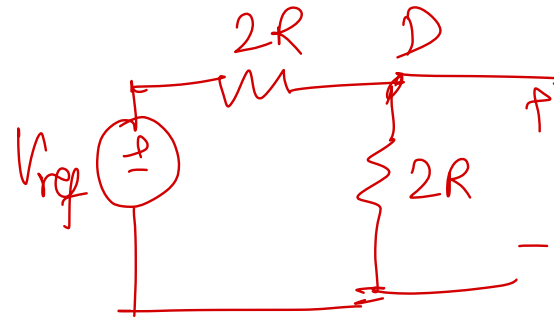
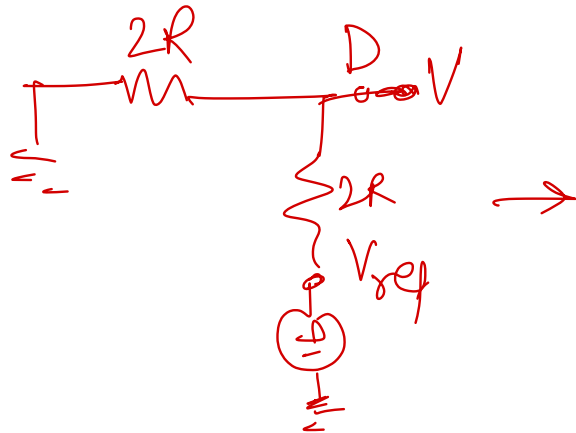
Where does the MSB and/or the LSB of the digital input go?

How to Analyse the R-2R ladder circuit?



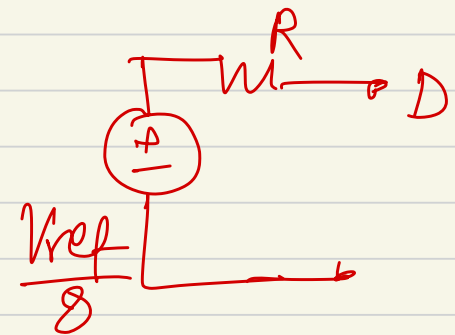
Case 1

$$V_D = +V_{ref}, V_A = V_B = V_C = 0$$

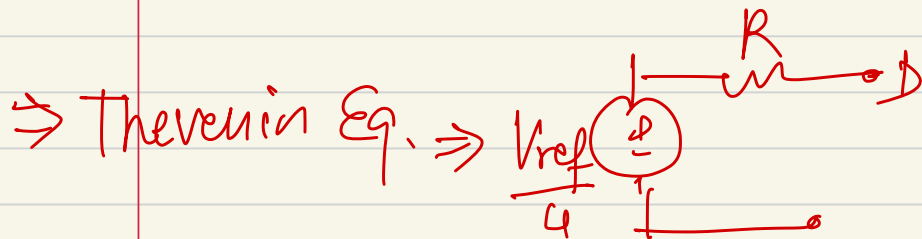


Case 3 $V_B = +V_{ref}$, $V_A = V_C = V_D = 0$

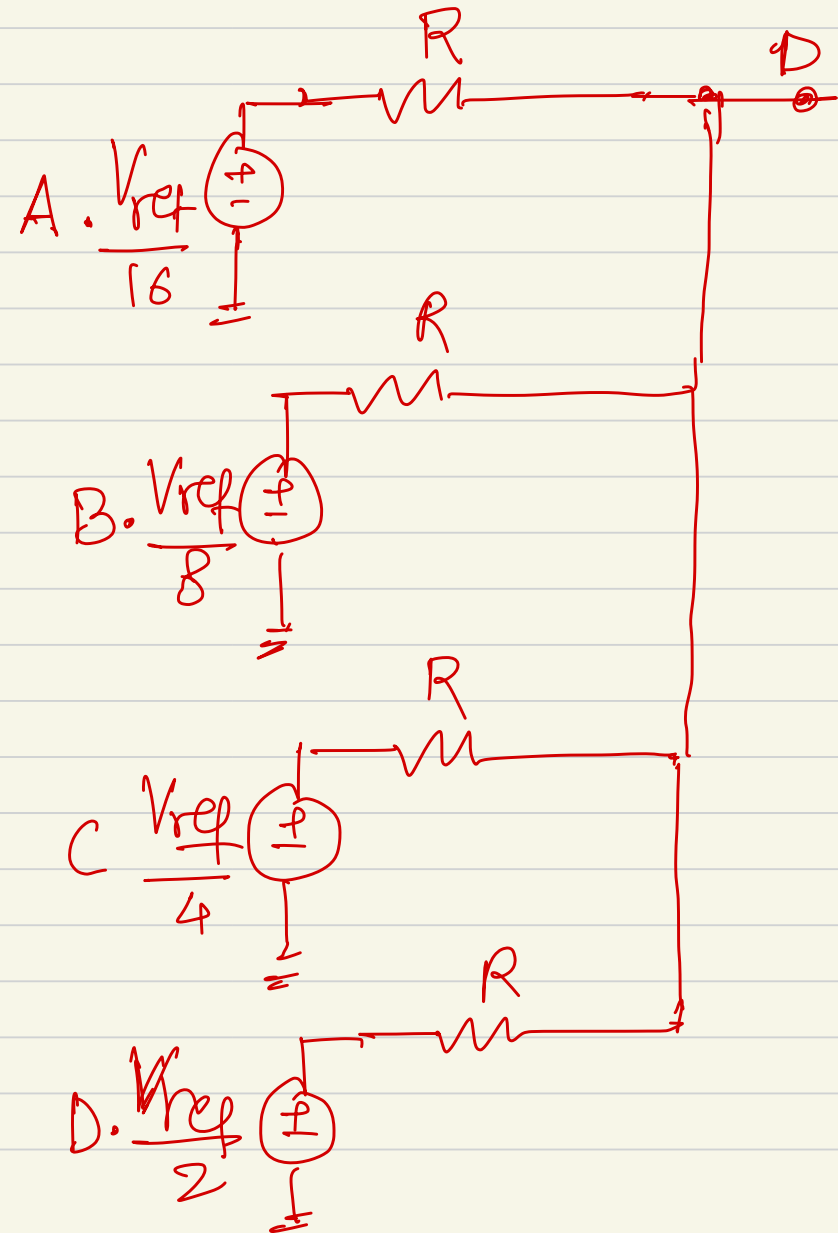
Thevenin
Equivalent $\Rightarrow$



Case 4 $V_C = +V_{ref}$, $V_A = V_B = V_D = 0$



Combined output
(superposition)



Analyse the full circuit (Tutorial 9:Q9)

5

